DA storage

Recover the bytes.
Keep their meaning outside.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A sealed storage locker: it preserves the package without deciding what the package means.

Why it matters: Every higher-level receipt depends on retrieving the same bytes. Keeping domain semantics outside DA makes the backend replaceable.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

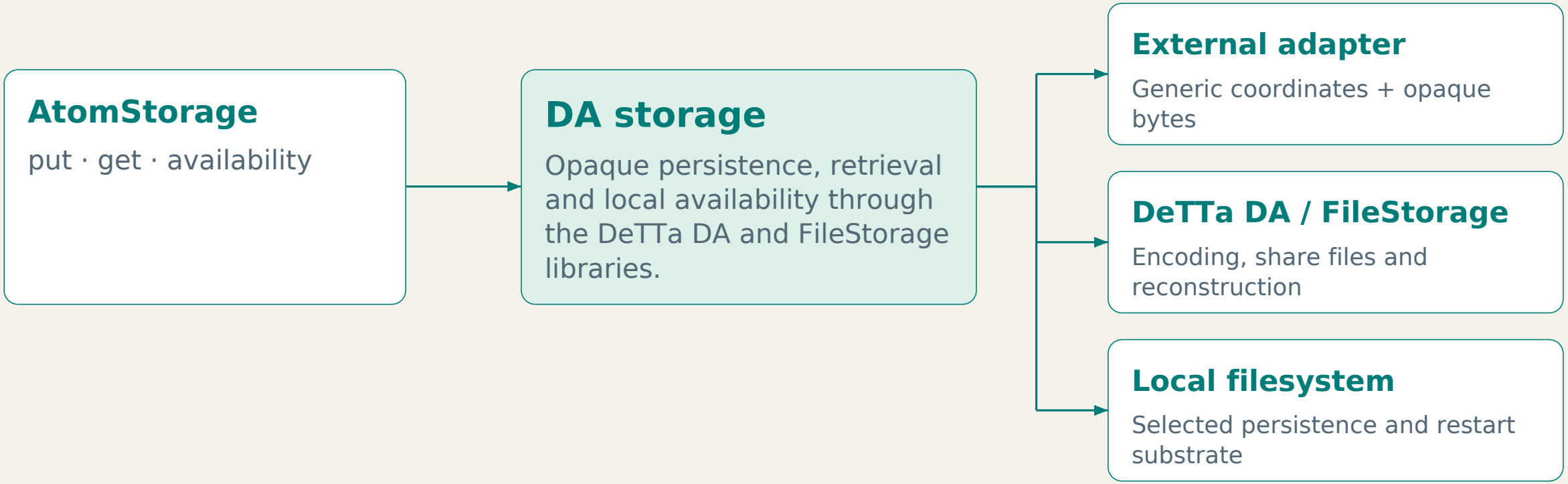
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Opaque persistence, retrieval and local availability through the DeTTa DA and FileStorage libraries.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Payload and share files

Local encoded bytes and reconstruction metadata

Persistence receipts

Evidence of the selected local storage behavior

OUTSIDE THIS RESPONSIBILITY

Application interpretation

No episode identity, belief, attention, job or query semantics

Distributed custody

No validator network, indefinite retention or global availability claim

opaque-local-rs/1 · receipt detta-local-rs/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
put	Persist opaque bytes at generic coordinates	Returns a provider receipt
get	Recover and verify the requested bytes	Unavailable/corrupt data fails explicitly
availability	Describe observed local availability	Does not prove future retention

Local files and trusted host; recovery does not imply distributed durability.

A worked example through the boundary

REAL LOCAL PROVIDER · generated payload and injected missing files

01 · STEP

Write

Store generated bytes via the real adapter.

02 · STEP

Remove

Delete the cached payload and two erasure shares.

03 · STEP

Recover

Reconstruct identical bytes from four remaining shares.

04 · STEP

Refuse

With a third share missing, report unavailable.

What the example establishes

Availability has a failure model. Preserve it in every receipt.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Opaque boundary

No domain identifiers or numerical meaning enter the DA core.

I2

Exact bytes

Successful retrieval must match the committed payload identity.

I3

Honest availability

Insufficient reconstruction material is not a successful read.

I4

Explicit evidence

A local restart receipt never becomes a custody certificate.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Missing cache + two shares	Four of six shares recover the exact payload.	Reconstruction test
Three missing shares / corruption	Fail explicitly; do not return fabricated bytes.	Negative storage test
Process restart	Reopen local files under the same profile.	Intact-file scope only

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4 + 2

Data + parity shares

Configured erasure profile, not measured throughput

2 lost

Shares recovered without cache

Observed reconstruction boundary

62 + 38

Upstream library tests

Clean DeTTa regression; overlapping with larger runs

No isolated read/write latency, throughput or durability benchmark is recorded. Share counts are a conformance result. Suite counts measure test execution, not performance.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary payload size, missing-share count and cold/warm cache; report read/write p50/p95, bytes stored and reconstruction CPU on fixed hardware.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Pinned DeTTa DA/FileStorage and external adapter run locally.
FIXTURE / LOCAL TEST POLICY	Generated payloads, deleted shares, corruption and injected process faults.
BLOCKED / UNSUPPORTED	No required current local gate blocked; historical dirty-source additions are outside the clean baseline.
UNRUN	Power loss, distributed custody, remote replicas, long-term retention and production load.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure storage tradeoffs	Controlled size/cache/failure matrix	Latency distribution and amplification per payload
Specify retention explicitly	New custody/retention contract and evidence	Expired retention cannot masquerade as availability
Explore another DA provider	Same opaque port and explicit profile	Equivalent byte/integrity conformance before adoption

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Availability has a failure model. Preserve it in every receipt.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py
```

Requires the pinned clean DeTTa source and built adapter; this is a suite command, not a microbenchmark.

`fusion/adapters/detta.py`
`adapters/detta/src/main.rs`
`docs/module-interfaces/m1_2-evidence.md`
`evidence/m5/legacy/m1_2-verification.json`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.